

About Simulation Scenario Standard - (SSS)

Simulation Must Actually Examine

A simulation can be technically sophisticated and still miss the conditions that matter most.

A model may be correct.

The environment may be controlled.

The input data may be suitable.

Yet the simulation may still provide weak insight if it examines only convenient, expected, or easily modeled conditions.

Simulation Scenario Standard (SSS) governs this problem.

It determines how materially relevant scenarios are identified, varied, combined, stressed, bounded, prioritized, and assessed for sufficient coverage before simulation execution begins.

Canonical Definition

Simulation Scenario Standard (SSS) defines the governance conditions under which simulation scenarios are identified, selected, varied, stressed, combined, bounded, and assessed for sufficient coverage so that the simulation examines expected conditions, materially relevant variations, edge cases, failures, uncertainty, stresses, transitions, and plausible alternative conditions capable of affecting simulation outcomes.

Why SSS Exists

Simulation quality is not determined only by how many simulation runs are performed.

A simulation may execute thousands or millions of times while still missing: critical edge cases, combined failures, degraded operating states, rare but severe events, transition failures, uncertainty ranges, environmental extremes, dependency failures, cascading events, or combinations that were never deliberately examined.

SSS therefore separates:

Simulation Volume

from:

Scenario Coverage

More simulation does not automatically mean better examination.

Governed Space

Simulation Scenario Design

SSS governs the condition space that exists between a prepared simulation basis and actual Simulation Execution.

It asks:

What must the simulation be asked to examine before its outputs can meaningfully represent the declared simulation inquiry?

Governance Flow

Governed Simulation Basis

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Scenario Requirements

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Baseline & Expected Conditions

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Scenario Variation

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Edge Cases

↓

Stress & Failure Conditions

↓

Scenario Combinations

↓

Scenario Coverage Assessment

↓

Coverage Gaps & Prioritization

↓

Governed Simulation Scenario Set



Simulation Execution From Simulation Capability to Simulation Examination

Before SSS, SIMULOS® establishes whether the simulation has a governed foundation.

SSS then converts that capability into an intentional examination plan.

The distinction is important:

The fact that a model can simulate a condition does not mean that condition was actually tested.

SSS therefore makes the Scenario Set itself a governed object.

Expected Conditions Are Only the Beginning

Nominal and expected operating conditions remain necessary.

They establish: baseline behavior, routine performance, expected operating states, and controlled comparison points.

But they are not sufficient where the Simulation Purpose includes materially relevant non-nominal behavior.

SSS therefore expands scenario design beyond the expected state.

Scenario Variation

Systems operate under variation.

Inputs change.

Environments change.

People behave differently.

Resources fluctuate.

Dependencies degrade.

Autonomous systems encounter situations not present in their nominal operating assumptions.

SSS governs deliberate variation across those conditions so that simulation behavior is examined across a meaningful condition space rather than at one convenient point.

Edge Cases

Some of the most important simulation failures occur outside the center of the operating distribution.

An autonomous vehicle may perform well in ordinary traffic but fail when: visibility is partially obstructed, multiple road users interact unexpectedly,

lane markings degrade, sensor confidence conflicts, or several individually manageable conditions occur together.

A spacecraft may behave correctly during nominal mission phases while a critical issue emerges only during: communication loss, degraded navigation, partial power availability, unusual thermal conditions, or a transition between mission modes.

SSS makes these conditions governable as Scenario Requirements rather than optional afterthoughts.

Stress Conditions

Stress scenarios intentionally push the simulation beyond normal conditions.

The objective is not to create arbitrary extreme tests.

The objective is to understand:

Where does behavior materially change?

Where do margins disappear?

Where do dependencies become critical?

Where does one degradation trigger another?

This is especially important for complex autonomous, aerospace, industrial, infrastructure, and mission systems.

Combined Conditions

Real failures are often not isolated.

A system may tolerate:

communication delay

and separately tolerate:

sensor degradation

while failing when:

communication delay + sensor degradation + reduced power

occur together.

SSS therefore governs materially relevant combinations rather than assuming that individual-condition testing establishes combined-condition coverage.

Transition Scenarios

Some systems are most vulnerable not in stable states but during transitions.

Examples include: launch, landing, docking, shutdown, restart, handover, human-to-autonomy control transfer, autonomy escalation, failover, recovery,

or operational-mode change.

SSS explicitly recognizes transition scenarios because:

safe State A + safe State B does not automatically mean a safe transition between A and B.

Failure and Degraded-State Scenarios

A simulation should not examine only complete success and complete failure.

Many operational problems occur in degraded states where the system still functions but under reduced capability.

Examples include: partial sensing, limited communication, degraded power, reduced computation, incomplete navigation, restricted human support, partial actuator capability, or weakened infrastructure.

These conditions are particularly important for autonomous and mission-critical systems because the system may continue operating while its decision environment has materially changed.

Rare but High-Consequence Conditions

Frequency alone does not determine scenario importance.

A condition may be rare but capable of producing catastrophic consequences.

SSS therefore supports consequence-aware scenario design.

The question is not only:

How often might this happen?

It is also:

What happens if it does?

Uncertainty Must Become Scenario Coverage

Upstream SIMULOS® standards may identify uncertainty in: assumptions, inputs, environment, parameters, or representation.

SSS converts materially relevant uncertainty into scenarios where necessary.

For example:

uncertain parameter range

may require:

low / central / high conditions

rather than one point estimate.

This prevents uncertainty from being documented upstream but ignored during

actual simulation.

Scenario Coverage

Scenario Coverage is one of the central outputs of SSS.

It asks whether the Scenario Set sufficiently examines the materially relevant dimensions of the simulation inquiry.

Coverage may include: nominal operation, variation, environments, populations, uncertainty, failures, transitions, stress, boundary conditions, rare events, dependencies, combinations, and recovery.

Coverage Is Not Scenario Count

Ten thousand scenarios may still provide poor coverage.

A smaller set may provide much stronger coverage if it deliberately spans the materially relevant condition space.

SSS therefore establishes:

Scenario Count ≠ Scenario Coverage

This is one of the most important practical distinctions in the standard.

Scenario Coverage Gaps

If a materially relevant condition cannot be simulated, SSS does not allow that absence to disappear.

The architecture records a:

Scenario Coverage Gap

or, where the simulation itself cannot represent the condition:

Scenario Capability Gap

This is valuable because it turns an unknown weakness into an explicit governance object.

Scenario Prioritization

For complex systems, exhaustive scenario enumeration may be impossible.

The number of possible combinations can become enormous.

SSS therefore allows governed prioritization based on: consequence, plausibility, uncertainty, system sensitivity, dependency concentration, boundary proximity, failure history, novelty, and Simulation Purpose.

The objective is not brute-force scenario generation.

It is governed selection of materially relevant scenario space.

Space and Mission Systems

Scenario governance has particularly strong application in space systems because many important conditions: cannot be reproduced economically in reality, may occur only once during a mission, are difficult or impossible to reproduce after deployment, involve interacting subsystem failures, depend upon changing environments, or occur during short but high-consequence mission phases.

A mission simulation may therefore need to examine not only:

nominal trajectory

but combinations such as:

communication degradation + navigation uncertainty + reduced power + thermal constraint during a mission transition.

SSS provides a structure through which such conditions can be identified, separated, combined, prioritized, and traced to actual simulation execution.

Autonomous Systems

Autonomous systems create a similar challenge.

Their behavior can depend upon combinations of: environment, sensor state, incomplete information, other agents, human behavior, communication, timing, uncertainty, and internal decision state.

SSS provides the scenario-governance layer through which these combinations can be deliberately examined rather than discovered only after real-world deployment.

Scenario Discovery

Simulation itself may reveal new scenarios.

An unexpected behavior may expose: a previously unknown dependency, a new failure sequence, a new interaction, or a previously unrecognized edge case.

SSS therefore treats Scenario Design as capable of evolving.

The lifecycle may become:

Scenario → Execution → Unexpected Behavior → New Scenario Requirement → Scenario Set Expansion

The original Scenario Set remains historically preserved.

No Outcome-Seeking Scenario Design

Scenario design can become biased if it focuses only on conditions likely to produce favorable outcomes.

SSS establishes a simple boundary:

Scenario design must remain capable of exposing unfavorable, contradictory, or unexpected system behavior where such behavior is materially relevant to the Simulation Purpose.

Relationship to SIDS

SIDS — Simulation Input & Data Standard establishes which information may enter the simulation.

SSS determines which combinations of those inputs, parameters, environmental states, and other conditions should actually be assembled into scenarios.

Therefore:

SIDS governs the information basis.

SSS governs the condition space examined.

Relationship to SMES

SMES — Simulation Model & Environment Standard establishes the model, environment, boundaries, parameters, and versioned simulation structure.

SSS configures that governed structure into deliberate scenario conditions.

Relationship to Simulation Assumptions

Scenario design may specifically test: assumption boundaries, uncertain assumptions, high-dependency assumptions, or conditions where assumptions may cease to remain applicable.

SSS does not govern assumptions themselves.

It determines when those assumptions should become the subject of deliberate simulated conditions.

Relationship to VALIDOS®

SSS does not determine whether scenario coverage is sufficient for a formal validation claim.

VALIDOS® may later consume scenario design, execution, and resulting evidence when evaluating validation requirements.

The boundary remains:

SSS governs what the simulation examines.

VALIDOS® governs what the resulting validation record can legitimately demonstrate.

Relationship to ORA™

ORA™ governs actual Operational Reality.

SSS may use real operational conditions, incidents, environments, or failure states as inputs to Scenario Requirements.

But the scenario remains a simulation object.

It is not Operational Reality itself.

Relationship to ASGA™ and AIG®

ASGA™ and AIG® may identify governance conditions relevant to autonomous-system or AI behavior.

Those conditions may become Scenario Requirements when the governed system is simulated.

For example:

autonomous escalation condition

may become:

simulation scenario requirement

without transferring autonomous-system governance into SSS.

Architectural Position

SSS is the seventh Parent Standard of SIMULOS® — Simulation Governance Architecture.

The progression is:

Simulation Object

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Simulation Purpose & Scope

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Reality Representation

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Simulation Assumptions

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Simulation Model & Environment

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Simulation Inputs & Data

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Simulation Scenario Design

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Simulation Execution

SSS therefore marks the transition from:

a simulation that is prepared

to:

a simulation that knows what it must actually examine.

Why It Matters

A simulation program can invest heavily in: better models, larger computing infrastructure, more data, higher fidelity, faster execution, and more simulation runs

while still leaving the most important conditions unexamined.

SSS provides a governance structure for identifying that gap.

That makes the standard useful not only for designing simulations but also for diagnosing existing simulation programs.

An organization may discover that its key bottleneck is not:

simulation capability

but:

scenario selection and scenario coverage.

Foundational Principle

A simulation cannot claim meaningful coverage merely because it executes successfully; it must deliberately examine the materially relevant conditions, variations, boundaries, uncertainties, failures, stresses, interactions, and alternatives capable of changing the behavior or conclusions of the simulation.

Canonical Closing Statement

Simulation Scenario Standard (SSS) governs the condition space that a simulation must deliberately examine before execution can meaningfully support

broader interpretation. By separating simulation volume from scenario coverage and governing expected conditions, variations, edge cases, boundaries, failures, degraded states, transitions, uncertainty, stress, combinations, prioritization, coverage gaps, scenario discovery, and scenario-to-execution traceability, SSS establishes a technology-neutral methodology for ensuring that simulation capability is converted into deliberate examination of the conditions most capable of changing system behavior or simulation conclusions.

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